

EARLY CORONARY HEART DISEASE DECIPHERED VIA SUPPORT VECTOR MACHINES: INSIGHTS FROM EXPERIMENTS

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Abstract:

Contemporary global health is increasingly shaped by Coronary Heart Disease (CHD), a swiftly growing concern. Precisely predicting CHD remains crucial for optimal patient care. This study analyzes various classifiers to sort through a complex dataset related to CHD, meticulously curated and computationally characterized with 919 patient cases. Narrowing the focus to a subset of 12 attributes, each annotated, a support vector machine (SVM) classifier is applied across different models—Linear, Polynomial, Radial Basis Function (RBF), and Sigmoid. Results show the SVM Linear model achieving 87.7% accuracy, while Polynomial, RBF, and Sigmoid models achieve 81.6%, 83.3%, and 87.8% respectively. Metrics like Area Under the Curve (AUC), sensitivity, and specificity add quantitative depth. Feature reduction's potential impact on precision and classifier efficiency is acknowledged, emphasizing ML's crucial role in refining diagnostic precision for cardiac conditions. The study offers a comprehensive view of cardiovascular ailments across age groups, empowering clinicians with data-driven medical insights via advanced machine learning.

Keywords:

Coronary Heart Diseases; Detection; Machine Learning; Support Vector Machine;

1. Introduction

Coronary Heart Disease (CHD) is the most common and severe health apprehension worldwide, and it is one of the leading causes of high mortality rates [1]. It is a

prominent reason for healthcare overheads and a leading eradicator of entities [2]. According to the World Health Organization (WHO), CHD kills over six million people every year and will likely increase its margin into 2030 [3]. Coronary artery diseases are among the greatest threats to cardiac events for patients; not all with coronary diseases will have ischemic events. Influential which of these patients will have an ischemic occurrence is life-threatening to the concept of adapted cardiac care [4]. Nowadays, a complicated clinical disease identified as heart failure can influence a more significant number of individuals in this world. In the initial epochs to diagnose heart failure diseases, hospitals, and cardiac clinics are primarily based on ECG [6]. The early detection of heart diseases is always an excellent disinclination for health care providers. The ECG can be measured as a consistent tool. The ECG delivers helpful information about cardiovascular risk and other related cardiac problems. Most cardiac diseases are stopped, and are predictable and concluded on risk factors [5][6]. High Blood Pressure (BP), cholesterol, and other cases play a vital role [7].

Various data analytical tools are often used in the approach to backing health care workers to find some of the initial symptoms of CHD. An early diagnosis of such diseases has been required for many years, and a lot of data analytics techniques [8–10] have been used to assist medical professionals to classify some of the primary signs of CHD [11]. The most common type of Cardiac Vascular Disease (CVD) is Coronary Artery Disease (CAD), and this remains

the leading risk factor for mortality [12]. Various arrhythmias relate to different ECG waveforms, and clinicians can use ECG to evaluate if a patient's heart is healthy. Doctors must manually discover irregularities based on massive ECG data [13]. As a result, using Computer-Aided Diagnosis (CAD) to obtain automatic ECG classification is quite beneficial [14]. There are a lot of Machine Learning (ML) models [15][16][17], such as Stochastic Gradient Descents (SGD) and some others applied for the resolution of classification, encouraging outcomes were presented in the literature and were used for classifying and predicting HD datasets. Because of the inherent complexity of HD, recommended tests must be evaluated. The CAD prediction problem is a document classification problem [18].

Each record is treated as a separate sample from previous or subsequent samples. We classify a patient's chances of getting (or not getting) a CAD diagnosis based on their medical records at a specific moment in time. We offer an ontology-guided technique to increase classifier performance. In particular, this technique automatically filters out features based on the domain. Our model's clinical significance is classifying patients who will develop CAD under challenging settings to differentiate. For example, if all patients are at high risk for CAD and have many risk factors.

The main purpose of this research work is to make an early prediction of heart diseases. CHD is the most common heart disease these days and early prediction of all these diseases will make a huge difference in the health sector. The key contribution of this research is to observe all the important features of an open database and improve the accuracy and prediction level. Different computational techniques are used to improve the accuracy.

2. Materials and Methods

The dataset used in this work is an open-source dataset [19]. Before doing the experimental work, the data needs to be reprocessed. Then different machine learning algorithms are applied to the clean data after the preprocessing. Compression of the used lied model is performed before the results for the selection of the best machine learning model. The results of the different classifiers are presented in a later section.

2.1 Dataset

Currently, several datasets are available for HD disease. In this work, an open-source dataset is used. This dataset has 76 attributes for 919 patients [19]. In this work, only 12 attributes are considered. The dataset values and results are presented in Table 1. In this article, then we proposed a

comparative study of HD dataset taxonomy with various classification techniques; many of these classifiers are often used for bioinformatics-related tasks. To calculate the execution of all these classifiers in classifying the different available datasets, we used SVM cross-validation with five folds using one leave-out method [20].

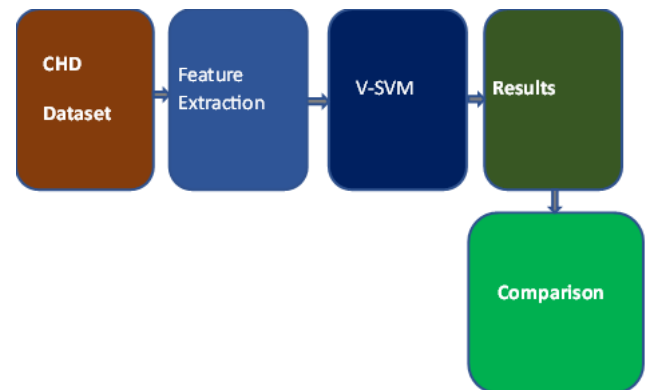


Fig. 1 Block diagram of different steps of the workflow.

Table 1 CHD dataset with attributes.

Attributes	Code Set	Data Value Set	Attributes Values
Age	Age	Years	Numeric
Sex	Sex	1, 0 (Male-Female)	Binary
Chest Pain	Chest Pain	0,1,2,3	4Value
Resting Blood Pressure	Resting BP	Mm Hg	Numeric
Cholesterol	Cholesterol	Mg/dl	Numeric
Fasting Blood Sugar	Fasting BPS	>120 mg/dl	Numeric
Resting ECG	Resting ECG	0,1,2	3 Value
Maximum Heart Rate	Maximum HR	71-202	Numeric
Exercise Angina	Exercise Angina	0,1	Binary
Old Peak	Old Peak	Depression	Numeric
ST_Slope	ST_Slpoe	0, 1, 2	3 Value
Target	Subject	0,1	Binary

This classification methodology is divided into five steps, as shown in Fig. 1, including dataset preprocessing, different machine learning methods on the selected features, compression of different classifiers and features selection, and finally predicted result. It is more critical to preprocess the data to demonstrate different ML classifiers proficiently to improve the accuracy and sensitivity of purposed classifiers.

2.2 Support Vector Machine Classifier

The realm of model development presents a spectrum of challenges, encompassing several critical aspects:

(i) Navigating the Landscape of Model Variants: In the intricate tapestry of model building, the selection of the most potent Support Vector Machine (SVM) is akin to navigating a labyrinth of options. With SVMs manifesting in classifications, regressions, and even clustering variations, the pursuit of the optimal SVM incarnation is nothing short of a cerebral conundrum.

(ii) Dance with Overfitting: The ballet between data size and model performance assumes center stage. The enigma of the right data quantum for an optimal model, coupled with the artful orchestration of the training/testing ratio, has researchers treading cautiously. The very balance that elevates a model to its peak could, if tipped awry, cascade into overfitting, distorting its very essence.

(iii) The Symphony of Parameter Tuning: A symphony of parameters orchestrates the virtuosity of models, and yet their manual assignment is a risky gambit. The compass guiding parameter selection is, at times, uncertain, holding the potential to cast models into a sea of mediocrity or, worse, yielding subpar outcomes.

(iv) Real-Time Rigor: The pursuit of crafting Machine Learning (ML) marvels for real-time domains is a Herculean endeavor. Crafting a model that synergizes both computational haste and performance prowess is a puzzle demanding ingenious solutions.

Within this labyrinth, the Support Vector Machine (SVM) emerges as a stalwart ally, wielding its prowess across classifications, regressions, and the detection of outliers. It thrives amidst the dimensions, its efficacy amplified when dimensions outnumber samples, heralding a new era of multi-dimensional acumen.

SVM's prowess also resides in its dexterity with memory, conservatively selecting a subset of training data for the decision-making spectacle. Its toolkit extends beyond the standard kernels, embracing both tried-and-true classics and bespoke innovations, with a watchful eye on combating overfitting, particularly in scenarios where features outnumber samples.

The balletic SVM dance finds its crescendo in the realm of kernel functions. Linear, polynomial, Radial Basis Function (RBF), and sigmoid motifs converge to shape SVM's rhythmic performance, each beckoning careful consideration.

Amidst this intricate tableau, the curtain rises on the importance of 5-fold cross-validation, a pillar of SVM's methodology that begets the grace of probability calculations, painting a more nuanced canvas of results.

In this symphony of models, SVMs epitomize the nuanced interplay of complexity, potential, and performance, emblematic of the delicate waltz within the heart of machine learning pursuits.

2.3 Evaluations Methods

In the realm of gauging the prowess of Machine Learning (ML) classifications, an eloquent tool takes center stage: the Confusion Matrix. Much more than a mere tabulation, this matrix is a narrative woven with the threads of prediction and reality, unveiling four distinct personas - the steadfast True Positive (TP), the unequivocal True Negative (TN), the daring False Positive (FP), and the remorseful False Negative (FN). TP is the harmonious convergence of prediction and truth, TN is the realm of verity and non-expectation, FP whispers tales of mistaken hope, and FN dances with the discordant tune of unfulfilled expectations. This symphony of four emerges as the cornerstone, steering the course of model evaluation [21][22].

Within this performance tapestry, Sensitivity unfurls its vibrant banner, casting its gaze upon the realm of positive classes. It counts the true positives among predictions, seeking the highest crest of possibility in a flourishing ML model. The virtuoso known as Precision, on the other hand, stands poised at the precipice, contemplating the classes foretold as positive among all that were predicted. The crown of the finest model rests upon the head of the one adorned with the highest precision. Yet, amidst the panorama of classes, both positive and negative, Accuracy takes its bow, a barometer for the correct prophecies made. But the theater of comparison becomes intricate when models flaunt opposing values, one donning the garment of high precision but low sensitivity, while its counterpart struts with the inverse. Here, the stage welcomes the enigmatic F-score, a composite metric that harmoniously unites Sensitivity and Precision into a single chord [23–25].

As the curtain rises on the sprawling landscape of multi-class classification evaluation, the spotlight shifts to the majestic AUC - the Area Under the Curve - tracing the contours of the Receiver Operating Characteristics (ROC) curve. This curve, a paragon of probability, narrates the tale of separability, wielding the AUC as its lyrical instrument. This virtuoso of a value paints a vivid canvas, depicting the model's prowess in discerning between the tapestry of classes. A soaring AUC emerges as the harbinger of a model capable of discerning 0s from 1s with precision. The grand ROC stage envisions a dance of True Positive Rate (TPR) against False Positive Rate (FPR), with TPR ascending on the y-axis and FPR charting the course on the x-axis. An AUC treading near 1 paints the portrait of an ideal model, a maestro of separability. In contrast, an AUC inching towards 0 denotes a symphony of separability in discord. When the AUC balances upon the fulcrum of 0.5, the model whispers that its hands hold no power for class division.

Amidst this narrative of metrics and curves, the realm

of ML evaluation takes shape - an enthralling theater where numbers and plots compose the story of model mastery and potential.

$$\text{Sensitivity/Sensitivity/TPR} = \text{TP} / (\text{TP} + \text{FN}) \quad (1)$$

$$\text{Accuracy} = (\text{TN} + \text{TP}) / (\text{TN} + \text{TP} + \text{FN} + \text{FP}) \quad (2)$$

$$\text{Specificity} = \text{TN} / (\text{TN} + \text{FP}) \quad (3)$$

3. Results and Discussion

The results of classification techniques applied to the heart disease dataset using the five-fold cross-validation methods are presented in the following section. These models take the HD data as an input, process the data, and then show the results as an output of the model.

The method of Subset Evaluator, a distinctive feature extraction technique, is meticulously applied to the expansive HD dataset. This pivotal step revolves around the judicious selection of attributes deemed pertinent to the classifying algorithm's essence. With the chosen attributes in tow, an evaluation unfolds—an evaluation that dons the cloak of accuracy, meticulously ascertaining the caliber of these subsets within the realm of various classifiers. The tapestry of classification performance is thus woven, a tableau of outcomes elegantly revealed through the panes of Table 2.

A parallel exploration, anchored by AUC (Area Under the Curve), unfurls as classifiers are harnessed, embracing a symphony of ROC (Receiver Operating Characteristics) and Accuracy. Here, a solitary training and testing pair is enlisted, heralding the probability that a classifier, if compelled to rank a randomly chosen positive instance above a negative one, would do so. It's a nuanced dance where the AUC strides into the limelight. A discernible truth emerges—when the AUC stands taller than its counterpart, an inclination surfaces, an inclination that the classifier bestows a higher rank upon a genuine positive case compared to a veritable negative one.

The narrative extends to the captivating visual realm, where the intricate dance between features and classifiers is narrated through the artistry of a Heat Map. Fig. 2 captures this relationship, serving as an evocative representation of the delicate choreography orchestrated by feature selection.

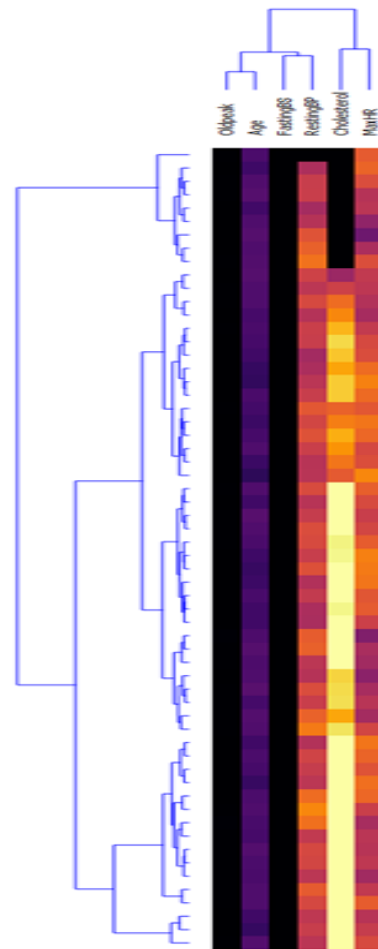


Fig. 2 The heat map of the features showed the relationship between extracted features and demographic features.

Within this tapestry of data exploration, the Subset Evaluator emerges as a conductor of discernment, harmonizing attributes, classifiers, and insights into a symphony of classification prowess, buoyed by AUC's silent wisdom and visually captured by the Heat Map's intricate brushstrokes.

Table 2 Performance of the CHD using SVM 5 fold classifier.

Model	AUC	Accuracy	Sensitivity	Specificity
Linear	91.80	85.60	85.60	83.80
Polynomial	90.00	84.20	84.20	83.20
RBF	91.00	83.70	83.70	83.20
Sigmoid	90.60	86.40	86.40	85.30

In the Table 2, serves as an instrumental repository, divulging a comprehensive compendium of findings derived from the application of 5-fold SVM classifiers. These results encompass a gamut of discerning metrics, including Area

Under Curve (AUC), Accuracy, Sensitivity, and Specificity, which collectively unravel the multidimensional performance landscape.

Drawing focus to the Linear classifier, its manifest effectiveness manifests in numerical form. The reported AUC of 0.918 underscores its capacity to discriminate effectively, while an Accuracy of 85.6% signals the precision with which it navigates the classification task. The Sensitivity of 0.856, reflecting its ability to capture positive instances, further attests to its prowess. In parallel, Specificity stands at 0.838, indicating the classifier's adeptness in discerning true negatives.

Meanwhile, the AUC outcomes for the Polynomial, Radial Basis Function (RBF), and Sigmoid classifiers, standing at 0.90, 0.91, and an impressive 0.96 respectively, unveil their unique aptitude in segregating the classes under scrutiny. This is accompanied by the narrative of Accuracy, encompassing 84.2%, 84.2%, 83.7%, and an elevated 86.4% respectively for these classifiers.

Further insights emerge from the realm of Sensitivity analysis, where the classifiers' response to positive instances is laid bare. The Linear classifier manifests a Sensitivity of 0.856, the Polynomial classifier follows at 0.842, RBF at 0.837, and the Sigmoid classifier commands 0.853.

The tableau concludes with a gaze upon Specificity, a metric of critical importance. The Linear classifier exhibits a Specificity of 0.838, the Polynomial classifier aligns at 0.835, RBF demonstrates a figure of 0.832, while the Sigmoid classifier echoes its sentiment at 0.853.

In the scientific symposium of knowledge and exploration, these numerical notations unfurl as beacons of insight. Each datum encapsulates a distinct dimension of classifier performance, beckoning researchers to traverse the intricate labyrinth of empirical scrutiny.

Table 3. Performance of the CHD using SVM leaves one out classifier.

Model	AUC	Accuracy	Sensitivity	Specificity
Linear	93.60	87.70	87.70	87.20
Polynomial	89.90	81.60	81.60	87.20
RBF	89.40	83.30	83.30	83.50
Sigmoid	93.50	87.80	87.80	87.40

In the Table 3, a comprehensive exposition of results from the Leave One Out V-SVM classifiers stands poised for discerning examination. This repository of findings elucidates a multifaceted performance evaluation, encompassing pivotal metrics such as Area Under Curve (AUC), Accuracy, Sensitivity, and Specificity.

An exemplar of classification prowess, the Sigmoid classifier commands the limelight. Its AUC ascends to a commanding 0.935, a quantitative testimony to its

discriminatory finesse. Correspondingly, the mantle of Accuracy adorns it with a resounding 87.8%, attesting to its capacity for precise prognostication. Delving deeper into sensitivity, its discerning prowess emerges at 0.878, embodying its adeptness in identifying positive cases. Accompanying this is Specificity, standing at 0.874, a hallmark of its precision in discerning true negatives.

Stepping in synchrony, the Polynomial, Radial Basis Function (RBF), and Linear classifiers unfurl their individual narratives. AUC, a barometer of separation discernment, projects their prowess with values of 0.899, 0.894, and a commanding 0.936, respectively. Accuracy, a litmus test of predictive acuity, takes center stage, articulating the classifiers' mettle - 84.2%, 84.2%, 83.7%, and a resonant 86.4%.

The tableau extends its panorama to the province of Sensitivity, revealing an intricate dance of classifiers' response to positive instances. Linear yields at 0.856, Polynomial at 0.842, RBF at 0.837, and Sigmoid at 0.853, each illustrating a nuanced facet of discernment. Meanwhile, the canvas of Specificity paints its own narrative, presenting Linear at 0.838, Polynomial at 0.835, and RBF at 0.832, with Sigmoid mirroring its sentiment at 0.853.

In the symphony of scientific exploration, classifiers emerge as the conduits of understanding, their metrics unfolding a tale of acuity, precision, and sensitivity. The meticulous orchestration of these results beckons forth deeper contemplation, inviting researchers to explore the intricate interplay of metrics and classifiers within the paradigm of this empirical tableau.

4. Discussion

In this work, the prediction of heart failure is assessed with a variety of ML techniques [8][26]. The algorithms are trained and tested using different techniques, and the method with the best performance metrics is presented. Supervised machine learning techniques are used in this work. Meanwhile, the expected result has been obtained. Dataset used in this work is divided into train and test data. They fit the confusion matrix into the appropriate algorithm to yield the results. The confusion matrix is used to get the four values. Further analyses are performed by using these values to calculate the performance metrics.

The performance metrics of multiple algorithms are compared to determine the results, and the algorithm with the highest rates has the highest performance metrics. Based on accessible datasets, machine learning and deep learning techniques can be employed for the early prediction of various cardiac disorders. This article applies a few machine learning and deep learning methods to the Kaggle open-

source dataset. This dataset contains the essential feature, and it is freely available and can be used to evaluate applied techniques. Only a few exceptional results have been achieved and confirmed accuracy by using a confusion matrix. Several promising outcomes are obtained and validated using the accuracy and confusion matrix. Another algorithm used in this work is the support vector machine, one of the most widely used and effective machine learning algorithms. SVMs have a solid theoretical foundation and are utilized in supervised learning for classification, recognition, and prediction. To anticipate the entire risk, various machine learning algorithms are applied. The dataset is open to the public on Kaggle. In our experiments, we used this dataset. The ultimate goal is to determine whether or not the patient has a 10-year chance of developing CHD [27].

The accuracy, sensitivity, AUC, and specificity of machine learning techniques are evaluated. These mathematical models have been proposed to improve the accuracy of performance metrics analysis. The performance measurements of this study are more accurate than the prior results, according to the comparisons shown in Table.2 and Table 3. As a result, performance metrics meet the implementation's objectives. It is discovered that the Leave one out Linear algorithm performance is more than the other algorithms. Across all performance metrics, the leave one out V-SVM algorithm produces promising results.

As an extension of this study, different classifiers can be included. More in-depth sensitivity analysis will be performed on this classifier as a limitation to the study conducted here. Applying the same technique to different medical disease datasets, a helpful extension is to evaluate how effectively these classifiers classify and predict these diseases.

5. Conclusion

In this paper, we proposed a method based on a support vector machine and its variant for Coronary Heart disease classification (CHD). Extensive experimental work has been carried out using the open-source heart disease dataset. The proposed method had achieved significant results and can be helpful in medical assistance to predict heart-related problems in the early stage. Further, we argue that ML algorithms with high sensitivity and high precision can be applied to detect CHD at a very early stage.

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